Structure of the Atom

The big idea

Dalton thought atoms were indivisible. Experiments around 1900 showed atoms contain even smaller particles: **electrons**, **protons** and **neutrons**. How these are arranged explains why elements behave the way they do.

Subatomic particles

Particle	Discovered by	Charge	Mass	Location
Electron (e^-)	J. J. Thomson	−1	About 1/2000 u	Outside nucleus, in shells
Proton (p^+)	E. Goldstein (canal rays)	+1	1 u	Nucleus
Neutron (n)	J. Chadwick (1932)	0	1 u	Nucleus

Models of the atom

- **Thomson**: atom is a positive sphere with electrons embedded, like seeds in a watermelon; overall neutral.
- **Rutherford's gold-foil experiment**: fast alpha particles fired at thin gold foil. Most passed straight through, some deflected, very few (1 in 12,000) bounced back. Conclusions: most of the atom is **empty space**; positive charge and nearly all the mass are in a tiny **nucleus**; electrons revolve around it. Problem: revolving electrons should lose energy and fall into the nucleus.

- **Bohr**: electrons move only in certain **discrete orbits (shells)**, K, L, M, N, without radiating energy.

Distribution of electrons

BOHR–BURY RULES

Maximum electrons in a shell = $2n^2$: K (n = 1) holds 2, L holds 8, M holds 18.
The outermost shell can hold at most 8 electrons.
Shells fill step by step: inner shells first.

Example configurations: Na (11) = 2, 8, 1; Cl (17) = 2, 8, 7; Ar (18) = 2, 8, 8.

Valency

Electrons in the outermost shell are **valence electrons**. Atoms tend to reach a full outer shell (an **octet**). **Valency** is the combining capacity: equal to the number of valence electrons (if 1–4), or 8 minus that number (if 5–7). Na has valency 1; Cl has valency 8 – 7 = 1; noble gases have 0.

Atomic number, mass number and isotopes

TWO NUMBERS

Atomic number Z = number of protons

Mass number A = protons + neutrons; so neutrons = A – Z

- **Isotopes**: same Z, different A (^{1}H, ^{2}H, ^{3}H; ^{12}C, ^{14}C). Same chemical properties. Uses: uranium isotope as nuclear fuel, cobalt isotope for cancer treatment, iodine isotope for goitre.
- **Isobars**: same A, different Z (^{40}Ca and ^{40}Ar).

CHLORINE IS ^{35}CL (75%) AND ^{37}CL (25%). FIND ITS AVERAGE ATOMIC MASS.

$35 \times 0.75 + 37 \times 0.25 = 26.25 + 9.25 =$ **35.5 u**.

Quick recap

- Electrons (–), protons (+) and neutrons (0); nucleus holds protons and neutrons.
- Rutherford: tiny dense nucleus; Bohr: fixed energy shells.
- $2n^2$ rule; valency from valence electrons.
- Isotopes share Z; isobars share A.

Practice questions

1. The neutron was discovered by
   (a) Thomson (b) Rutherford (c) Chadwick (d) Bohr
2. The maximum number of electrons in the M shell is
   (a) 8 (b) 18 (c) 32 (d) 2
3. The electronic configuration of sodium (Z = 11) is
   (a) 2, 9 (b) 2, 8, 1 (c) 2, 7, 2 (d) 8, 2, 1
4. An atom has 17 protons and 18 neutrons. Its mass number is
   (a) 17 (b) 18 (c) 35 (d) 1
5. Isotopes of an element have the same
   (a) atomic number (b) mass number (c) number of neutrons (d) physical properties
6. In Rutherford's experiment, most alpha particles passed straight through because
   (a) the nucleus is very large (b) most of the atom is empty space (c) electrons attracted them (d) gold is soft

Answers: 1. c 2. b 3. b 4. c 5. a 6. b